# Seth Temple

MACHINE LEARNING AND STATISTICS RESEARCHER

**Los Alamos, NM | sethtem@umich.edu | linkedin.com/in/sdtemple**

## Education

**University of Washington, PhD, Statistics** **2024**

**University of Washington, MS, Statistics** **2022**

**University of Oregon, BS, Mathematics & Computer Science (*summa cum laude*)** **2018**

## Experience

**Los Alamos National Laboratory | Postdoctoral Research Associate** **07.2026 – Present**

- Designing neural network estimators that train without gradients as open source software
- Benchmarking against CNNs and large tabular models in regression & image classification tasks
- Exploring strategies from random matrix theory to initialize and sparsify neural network weights

**University of Michigan | Postdoctoral Research Fellow** **09.2024 – 06.2026**

- Designed a hypothesis testing method to protect against false discoveries in case-control studies with the Alzheimer's Disease Sequencing Project consortia
- Proposed Bayesian and frequentist estimators of measles infection rates with partial case data
- Performed positive selection analysis for an international consortium on chicken genetics
- Built and taught an AI models workshop on CNN and transformer classifiers for acoustics data

**University of Washington | Graduate Student Researcher** **09.2019 – 08.2024**

- Performed 5 whole-genome sequence analyses of 3 ancestry groups in TOPMed & UK Biobank to determine regions indicative of selection or technical artifacts
- Created a suite of 4 methods to pinpoint and model the effects of advantageous genetic variants
- Developed a fast and scalable algorithm to simulate data for biobank-scale inferences
- Proved central limit theorems in population genetics relevant to downstream analyses

**Fred Hutch Cancer Research Center | Graduate Student Researcher** **06.2023 – 12.2023**

- Designed a time series method to detect and cluster rapidly evolving SARS-CoV-2 variants
- Collaborated on an international project modeling genomic patterns of synonymous mutations

**Los Alamos National Laboratory | Graduate Student Researcher** **06.2020 – 10.2020**

- Developed and deployed a Bayesian spatial model for mosquito-borne epidemiology
- Implemented spatial ML method to predict and visualize continent-wide mosquito prevalence

**Liberty Mutual Insurance | Actuarial Assistant** **07.2018 – 08.2019**

**University of Oregon | Research Assistant** **01.2018 – 06.2018**

## Skills

- Python (10+ years), R (8+ years), Bash, git, SQL, and C++ programming
- High-performance computing with slurm (6+ years)
- Deep learning with (CNNs, LSTMs, ViT)
  - PyTorch, Transformers, TensorFlow, scikit-learn
  - RayTune and Optuna for hyperoptimization
- Methodology development across disciplines
  - Statistical genetics, genomics, and epidemiology
  - Bayesian inference and hierarchical models
  - Spatial data science and time series
- Algorithm, software, and bioinformatic workflow development
- Science communication and technical writing
- Project planning and execution
- Statistical consulting

## Software

Individual

- github.com/sdtemple/isweep: Python package and 5 automated pipelines for genetic analyses (4+ years of continuous software development)
- github.com/sdtemple/peirrs & github.com/sdtemple/pblas: R packages for estimators in epidemics

Contributor:

- github.com/lanl/GFDL: training neural networks without backpropagation

Tutorials:

- github.com/sdtemple/zootopia3 & huggingface.co/sdtemple/color-prediction-model: fitting NNs
- github.com/sdtemple/your-first-model: hack-a-thon setup for deep learning on acoustics data
- github.com/sdtemple/how-to-use-ai: using AI coding assistance and agents

## Awards

- Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship [UM]
- National Defense Science and Engineering Graduate Fellowship (2% selection rate) [UW]
- Z.W. Birnbaum award for best PhD candidate exam [UW]
- NIH Traineeship in Statistical Genetics [UW]
- Phi Beta Kappa academic honor society [UO]

## Teaching

- Lead Instructor for Graduate Statistics Course in Statistical Genetics [UW]
- Teaching Assistant: Intro to Machine Learning, Experimental Design, Applied Regression [UW]
- Directed Reading Program Mentor for 4 students [UW]
- College Math Tutor for 4 years [UO] and K-12 Math Tutor for 1 year [Seattle Public Libraries]

## Publications

7 first-author and 5 co-author publications on statistical methods in biological sciences,
3 of which are published in the 8th most cited Genetics and Genomics journal

1. Temple, S.D., & Terhorst, J. Pair-based estimators of infection and removal rates in partially observed stochastic epidemic models. arXiv (2026).
2. Temple S.D., et. al. Multiple-testing corrections in case-control studies using identity-by-descent segments. The American Journal of Human Genetics (2026).
3. Temple, S.D., & S.R. Browning. Multiple-testing corrections in selection scans using identity-by-descent segments. The American Journal of Human Genetics (2025).
4. Temple, S.D., & Thompson, E.A. Identity-by-descent segments in large samples. Theoretical Population Biology (2025).
5. Temple, S.D., Browning, S.R., & Thompson, E.A. Fast simulation of identity-by-descent segments. Bulletin of Mathematical Biology (2025).
6. Temple, S.D., Waples, R.K., & S.R. Browning. Modeling recent positive selection using identity-by-descent segments. The American Journal of Human Genetics (2024).
7. Temple, S.D., Manore, C.A., & Kaufeld, K.A. Bayesian time-varying occupancy model for West Nile virus in Ontario, Canada. Stochastic Environmental Research and Risk Assessment (2022).

8. Browning, S.R., Temple, S.D., & Browning, B.L. Local ancestry inference with poorly-matched reference panels. PLoS Genetics (2026).
9. Haddox, H.K., et. al. Clonal interference and changing selective pressures shape the escape of SARS-CoV-2 from hundreds of antibodies. Virus Evolution (2026).
10. Haddox, H.K., Angehrn, G., Sesta, L., Jennings-Shaffer, C., Temple, S.D., et.al. SARS-CoV-2's mutation rate is highly variable between sites and is influenced by sequence context, genomic region, and RNA structure. Nucleic Acids Research (2025).
11. Horimoto, A.R.V.R., et. al. Admixture mapping implicates 13q33.3 as ancestry-of-origin locus for Alzheimer disease in Hispanic and Latino populations. HGG Advances (2023).
12. Gorris, M.E., Bartlow, A.W., Temple, S.D., et. al. Updated distribution maps of predominant Culex mosquitoes across the Americas. Parasites and Vectors (2021).
13. Chapman, N. H., Xue, D., Temple, S.D., et. al. Exploring local-ancestry effects on the relationship between APOE variation and lipids and inflammatory markers. To be submitted to The American Journal of Human Genetics (Autumn 2026).
14. Fiddaman, S., et. al. Quantifying 2,000 years of selection in chickens using ancient DNA and functional genomics. To be submitted to Cell (Autumn 2026).

15. Temple, S.D. Statistical Inference Using Identity-by-Descent Segments: Perspectives on Recent Positive Selection. University of Washington (Ph.D. thesis, 2024).
16. Temple, S.D. The Tweedie Index Parameter and Its Estimator: An Introduction with Applications to Actuarial Ratemaking. University of Oregon (Bachelor of Science thesis, 2018).